Python-Day-18

Python operators:

→

→ objects → perform a operation → syntax ⓪]

① operators : syntax → operations perform

② operands : values on which we perform operations.

① addition

a = 10 → assignment → a → obj → @

b = 10 → operator

→ print(a+b) # 20 → addition

operands

Arithmentic operators:

→ arithmetic operations → in b/n operands

→

① addinen

10 + 10 → addition

a = 10 → obj

b = 10 → obj

print(a+b) → + → sum pyth.ey

→

② substation (-)

a = 10

b = 10

print (a-b) #0

multiplication(*)

x = 10
y = 2
print(x*y) #20

Divison(/)

x = ~~10~~
y = ~~2~~
print(x/y) 5

% → modulus → return reminder

x = 10
y = 2
print(x%y) #0

(**) → exponential

x^n 2×2 = 4

2**2

2^2

x=2
y = 2
print(x**y)

→ floor Division (//)

a= 10
b=3
print(a//b) 3

3.33

Assignment operator : → Normal assignment

Compound assignment

→ values → variables

→

Normal assignment :

a = 10
b = 10
print(a+b)

a = 10
b = 10
a+=b → a = a+b

Compound assignment

a = 10
b += a
print(b)

a = 10 → obj
b = 10
a += b → new obj
print(a)

a = 10
print(id(a)) → id → differ

b = 10

10 + 10

a += b → a = a + b
= 20 → id →

print(id(a))
print(a) => 20

=

= → python.exe → assign

@ → 0

bit → binary

-=
+= arithmetic
*= compound assignment
**=
/=
//=

& &=
|
&=
|=

Bitwise compound assignment
^=

a = 5 → 0 1 0 1
b = 2 → 0 0 1 0
a &= b
print(a)

8 4 2 1
0 1 0 1
0 0 1 0
0 0 0 0 = 0

1 == 1

→ walrus (:=) (:=)

→ expression → condition, loop state

if x = 5: → error → python.exe → syntax error
print(x)

→ in expression we are able to assign a value

→ if x := 5: → 5
print(x)

Comparison operators:-

-> Relational operators
-> it used compare the the values by using variables.
-> it return True/false

-> == -> equal -> two values same or not

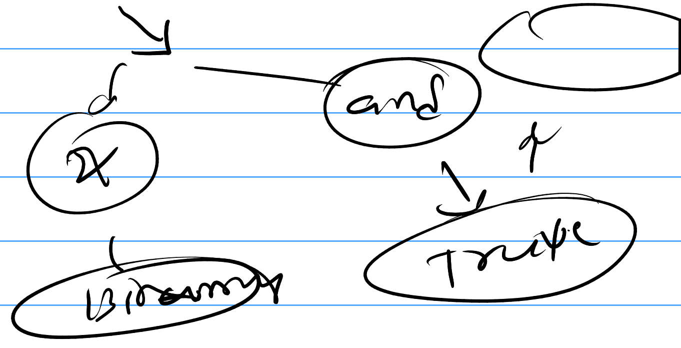
x = 10
y = 10
print(x==y) # True//

-> Not equal -> != print(x!=y) # False

-> > -> greter than -> print(x>y) # false
-> < -> Less than -> print(x<y) # false
-> >= -> greter than equal -> print(x>=y) # True
-> <= -> Less than equal -> print(x<=y) # True //

-> it can able compare in chain

x = 5
print(1<x<10) # True
print(1<x and x<6) # True



Logical operators:
-> Bolean values -> True/false

vote eligible

18 = , 60

-> and -> True, True

age = int(input("enter age: "))

if age>18 and age<60:
    print("vote eligible")

-> combine the codition and evalute them pritat result ->

T8 -> True

20

Not

A	B
T	F
F	T

80>1

True

True T

print(not(10))

False

① And or Not

x	y	x and y	x or y	Not(x)	Not(y)
T	T	T	T	F	F
T	F	F	T	F	T
F	T	F	T	T	F
F	F	F	F	T	T

T -> F
F -> T

0 -> false, [], (), None
1 -> True

-> false

x = 0
y = 10

False True

or

if x and y:
    print(y, x)

=> empty list

=> False

if False and True

10, 0

if True and False

print

X